Pro Memoria: Skip-Based State Transmission for Long-Horizon Agent Handoffs

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Abstract. Sequential agent workflows that hand work between fresh workers face a growing transmission problem: if every worker receives the accumulated conversational history of its predecessors, transmitted context grows with each handoff, and cumulative token consumption grows faster than the task itself. This paper presents Pro Memoria (PM-1), a handoff architecture organized around a single principle: *skip, don't compress.* Rather than retransmitting accumulated history in a smaller representation, PM-1 does not retransmit it at all; each worker receives a bounded continuation-state packet containing the state required to continue. The architecture is designed to be encoding-agnostic: the original implementation used Morse, and the benchmark reported here uses a PM-1-shaped JSON packet.

We evaluate the architecture across 30,100 completed API-backed handoffs in two sequential experiments. A 100-hop state-survival experiment (4,000 calls, 10 chains, 4 conditions) measured 100% state integrity, 100% digest continuity, and zero protocol-induced state drift. A token/context scaling experiment (26,100 calls) compared a bounded PM-1 state packet against an accumulating conversational transcript over horizons 10–250. The packet condition maintained bounded per-hop context of approximately 78–79 tokens and exhibited approximately linear cumulative token growth (linear fit $R^2 = 0.999998$), while the conversational condition exhibited strongly super-linear cumulative growth (quadratic fit $R^2 = 0.999999$) — a 90.43% cumulative-token reduction relative to the accumulating conversational condition at horizon 250. In the packet condition, 99.98% of hops produced successfully formatted worker responses and 100% preserved state integrity and digest continuity; the three formatting failures produced no state divergence.

These results support claims about sequential state survival and the transmission economics of skip-based handoff within this benchmark. They do not address semantic knowledge transmission, institutional memory, or cross-model generalization, which are out of scope and are the subject of future work.

Keywords: agent handoff; context management; state transmission; token economics; long-horizon agents

1. Introduction

Long-running agentic systems decompose tasks into steps, each executed by a fresh worker, with results flowing to the next worker. A common implementation passes the accumulated conversation — every prior state, action, and reasoning trace — to each new worker. As the horizon grows, this accumulated context occupies more of the model's context window, cumulative consumption grows faster than the task, and the transmitted context eventually approaches the window limit. Empirical evidence shows that language models do not robustly use arbitrarily long contexts even when they fit [Liu et al., 2024], so an accumulating transcript is not merely expensive — it may also degrade the reliability of the information a worker actually needs. PM-1 addresses one architectural mechanism that may contribute to long-horizon context pollution: the unnecessary retransmission of conversational history across handoffs.

This paper examines an alternative design. Pro Memoria (PM-1) is a handoff architecture whose organizing principle is **skip, don't compress**: the next worker does not receive the accumulated history of how prior workers arrived at the current situation; it receives a bounded packet containing the state required to continue. The distinction from compression is fundamental. Compression transmits the same accumulated information in a smaller representation; PM-1 changes what crosses the handoff boundary, so that known or unnecessary history is not transmitted at all.

The research question is:

> For a sequential task executed by fresh workers, how does the cumulative cost of transmitted context grow with horizon under skip-based bounded state transmission compared with accumulating conversational retransmission?

We evaluate this with two sequential experiments. V0.4a asks whether PM-1 state can survive repeated sequential handoffs. V0.5 asks what transmission and context cost is incurred by repeatedly handing off state while avoiding accumulated conversational history. Together they provide 30,100 completed API-backed handoffs.

Contributions.

1. **An architectural principle.** Skip-based state transmission decouples per-handoff transmission cost from accumulated history length.
2. **A protocol definition.** PM-1 specifies a bounded continuation-state packet (`sections → records → payload`), a fresh-worker handoff boundary, and digest-continuity integrity semantics.
3. **An empirical evaluation.** Two sequential experiments measure state survival over 100 handoffs and the cumulative token/context consequences of skip-based handoff over horizons 10–250.

We do not claim that PM-1 invented state handoff, structured state, or agent memory — structured state sharing has long precedent in blackboard architectures [Hayes-Roth, 1985; Nii, 1986]. The claim is narrower: PM-1 formalizes the handoff boundary as a selective transmission boundary and empirically evaluates bounded continuation-state transmission against accumulating conversational history across repeated fresh-worker handoffs.

2. Problem Definition

Consider a sequential task of horizon **N** executed by fresh workers: at each step **i**, a worker receives transmitted context, produces an action, and is destroyed; the next worker receives new context for step **i+1**. The task's **continuation state** is the minimal set of variables the worker needs to determine the correct next action.

Two transmission strategies are compared:

- **Condition P (bounded state packet).** Each worker receives a bounded continuation-state packet — the semantic state plus minimal identifying fields — and an immutable task specification. Packet size is independent of **N**.
- **Condition C (accumulating conversation).** Each worker receives the chronological transcript of all prior (state ⟶ action) entries plus a current-state line, and the same immutable task specification. The transcript grows with every hop.

The measured quantities are per-hop transmitted context and cumulative transmitted tokens as functions of **N**.

The problem is deliberately scoped to **state transmission**, not knowledge transmission. The continuation state in this benchmark is a small deterministic portfolio state; the task is a vehicle for measuring handoff economics, and the paper makes no claim about trading.

Design goals. Three goals govern the design: (1) **boundedness** — per-handoff transmission must not grow with horizon; (2) **continuation** — a fresh worker receiving only the packet and the task specification must determine the correct next action; (3) **integrity** — the continuation state must survive handoff intact, with a verifiable digest. These goals are orthogonal to compression: a compressed transcript still grows with history; a skip-based packet does not.

3. Pro Memoria Architecture

The architecture separates four concerns:

1. **Canonical state.** The continuation state the next worker needs, materialized as records in a `sections → records → payload` packet shape.
2. **Handoff boundary.** Each worker runs in a fresh session with no access to prior sessions, the state directory, hidden truth, or the oracle.
3. **Transmission.** The bounded packet plus the immutable task specification is the only task-relevant context the worker receives.

4. **Validation.** An independent oracle validates actions against hidden truth; a deterministic ledger applies transitions; digests verify state continuity.

3.1 Skip semantics

The core operation is **skipping**, not **shrinking**:

- Known or unnecessary history is not retransmitted.
- The receiving worker receives the state required to continue.
- The accumulated transcript is dropped at each handoff boundary, not compressed and forwarded.

Two clarifications position this correctly. First, **skip** is not merely **choosing a smaller state representation**: selecting a compact state is necessary but not sufficient. The mechanism claim is that accumulated conversational history does not cross the handoff boundary at all, which the P-vs-C comparison isolates — both conditions solve the same task, and only the transmission of prior history differs. Second, structured state itself is not novel (blackboard architectures transmit state rather than interaction history [Hayes-Roth, 1985; Nii, 1986]); the contribution is the formalized selective handoff boundary for fresh LLM workers and the measured transmission consequences.

The economic consequence: because the packet size does not depend on history length, cumulative transmission grows with the number of hops rather than with the product of hops and accumulated history. This is the mechanism behind the measured scaling separation in Section 6.

3.2 PM-1 packet model

A PM-1-shaped state packet is a JSON document of the form `sections → records → payload`. In this benchmark the packet contains a single `project_state` section whose payload carries the continuation state: `position_qty`, `target_signal`, `cash_cents`, `instrument`, `price_cents`, `logical_tick`, and tags. The state digest is computed over the semantic state fields (`position_qty`, `target_signal`) so that digest continuity measures continuation state rather than experiment identity.

3.3 Encoding

PM-1 is defined by what is transmitted and what is skipped, not by any particular encoding. Morse is the original encoding implementation and part of the architecture's provenance. V0.5 uses a PM-1-shaped JSON packet. The successful substitution of JSON for the original Morse encoding demonstrates that Morse is not required by the architecture; broader encoding agnosticism remains to be evaluated.

3.4 Scope of the benchmark implementation

V0.5 exercises a PM-1-shaped payload subset rather than every feature of the original implementation:

1. The packet is a payload-shaped subset; full envelope semantics were not exercised.
2. Selection was intentionally trivial: one `project_state` record per hop.
3. The benchmark transmitted a bounded full-state snapshot; field-level delta selection was not tested.
4. The packet used JSON rather than Morse.

These are scope boundaries, not defects; the paper does not claim V0.5 tested intelligent memory selection.

4. Experimental Design

4.1 Task

The benchmark uses a deterministic synthetic portfolio/state task. It is not a trading claim. It exists because it provides deterministic state, a deterministic oracle, controlled transitions, measurable state divergence, and repeatable long-horizon handoffs. The scientific object is the handoff architecture.

4.2 Model and configuration

Single model (deepseek-v4-flash), temperature 0.0, maximum output tokens 2048, identical between P and C. Workers run in fresh contexts; the condition, horizon, scenario, and trial are hidden from the worker; there is no oracle leakage, no expected-action leakage, and no token-count leakage. No heterogeneous-model or cross-provider validation is claimed.

4.3 Token and context measurement

Provider-reported `usage` (prompt/completion tokens) is captured per hop. When the provider omits usage, a deterministic `chars/4` estimate is used and reported explicitly. Cumulative input/output/total tokens are tracked per hop and per chain; token-completeness was verified at 26,100 / 26,100 records. Transmitted context size per hop is recorded in characters (exact) and estimated tokens (`chars/4`): PM-1 packet size per P hop, conversation history size per C hop.

4.4 Horizons and scaling analysis

Horizons 10, 25, 50, 100, and 250. H500 and H1000 were registered but not executed; they did not fail, and no results are extrapolated for them. All executed horizons evaluate a prefix of the same master chain specification. Linear and quadratic least-squares fits (pure-python normal equations) are fit to cumulative tokens versus horizon. Growth is reported as measured approximately-linear versus strongly super-linear over the observed horizons H10–H250; the quadratic fit is a characterization of the measured data, not an asymptotic proof.

4.5 Reliability metrics and acceptance criteria

Reliability metrics: action correctness, received-state correctness, oracle-state correctness, state integrity (divergence == 0), digest continuity, semantic drift, handoff survival, chain survival, parse errors, and failure-taxonomy counts, per condition and horizon.

Acceptance criteria (registered before API calls): reliability equivalence between P and C within ±5 pp at each horizon; P cumulative-context scaling linear $R^2 \geq 0.95$; C cumulative-context scaling super-linear (quadratic R^2 > linear R^2); PM-1 boundedness (P maximum transmitted tokens ≤ 2× from H10 to H1000); C context growth (> 5× from H10 to H1000); zero metadata leakage; zero oracle leakage; deterministic reproducibility; token completeness.

5. Results

5.1 V0.4a — state survival across 100 sequential handoffs

Question: can PM-1 state survive repeated sequential handoffs?

Design: 4,000 API calls; 100-hop sequential handoffs; 10 chains × 100 hops × 4 conditions (H-full, H-direct, H-corrupt, H-recover).

Metric	Result
100-hop completion	100% in every condition
State integrity (divergence == 0)	100% in every condition
Digest continuity	100% (H-full, H-direct)
Protocol-induced state drift	0
H-full handoff success	999 / 1,000
H-full failure	1 model PARSE_ERROR (chain-05, hop 24), no state consequence

Table 1: V0.4a results summary (4,000 calls, 10 chains, 100 hops, 4 conditions).

The single H-full failure was a worker output-format failure, not a protocol failure; it moved chain survival to 9/10 (−10 pp) but produced no state divergence. H-corrupt also recorded one model PARSE_ERROR (chain-08, hop 40) at a non-corruption hop with no state consequence; H-direct and H-recover completed 1,000/1,000.

5.2 V0.5 — token/context scaling over horizons 10–250

Question: what is the transmission/context cost of repeatedly handing off state while avoiding accumulated conversational history?

Design: 26,100 completed API calls; horizons 10, 25, 50, 100, 250; 10 scenarios × 3 trials × 2 conditions (P, C). Same deterministic task, same model, same configuration; the only independent variable is how state/history is transmitted.

Dataset reconciliation: expected and actual persisted hop records 26,100 / 26,100; checkpoint match true; duplicate call IDs 0; missing call IDs 0; token usage records 26,100 / 26,100; recomputed actual-digest mismatches 0; expected-vs-actual digest mismatches 2 (both in C at H100 — see Section 5.5).

5.3 Cumulative token totals

Horizon	P total (tok)	C total (tok)	C/P	Reduction 1–P/C
10	149,590	168,676	1.128	11.32%
25	370,703	633,117	1.708	41.45%
50	737,870	1,957,375	2.653	62.30%
100	1,474,888	6,819,750	4.624	78.37%
250	3,672,305	38,366,105	10.447	90.43%

Table 2: Cumulative token totals by horizon and condition.

At H250, the PM-1 packet condition consumed 3,672,305 cumulative tokens versus 38,366,105 for the conversational condition: a **90.43% cumulative-token reduction relative to the accumulating conversational condition at H250**. This is a difference between two handoff strategies within this benchmark; it is not a compression ratio applied to history, and it is not memory compression.

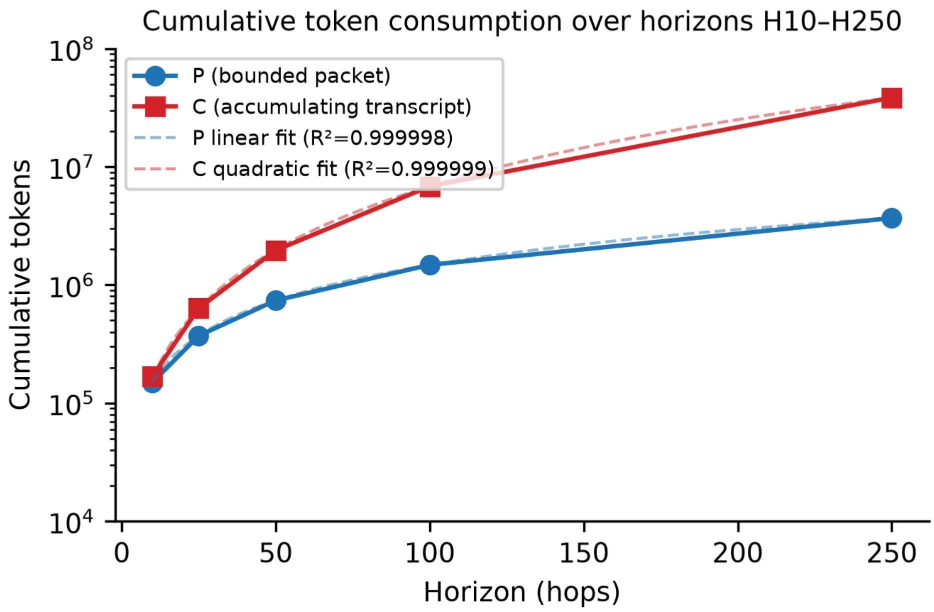

Figure 1: Cumulative token consumption over horizons H10–H250, with fitted curves.

5.4 Per-hop transmitted context

Horizon	Cond	Min tok	Max tok	Mean tok	Min chars	Max chars	Mean chars
10	P	78	79	78.67	314	319	317.04
10	C	11	356	161.81	46	1,427	648.78
25	P	78	79	78.63	314	319	316.84
25	C	11	881	396.61	46	3,526	1,588.05
50	P	78	79	78.61	314	319	316.77
50	C	11	1,833	788.28	46	7,332	3,154.72
100	P	78	79	78.61	314	319	316.73
100	C	11	3,618	1,609.30	46	14,474	6,438.74
250	P	78	79	78.60	314	319	316.71
250	C	11	8,864	3,977.78	46	35,457	15,912.62

Table 3: Per-hop transmitted context by horizon and condition.

P per-hop transmitted context remained bounded at approximately 78–79 tokens / 314–319 characters across all horizons. C grew continuously, reaching a maximum observed per-hop transmitted context of approximately 8,864 tokens / 35,457 characters at H250. The provider genuinely consumed the growing C context: cumulative C totals are computed from provider-reported prompt tokens, not from the `chars/4` heuristic.

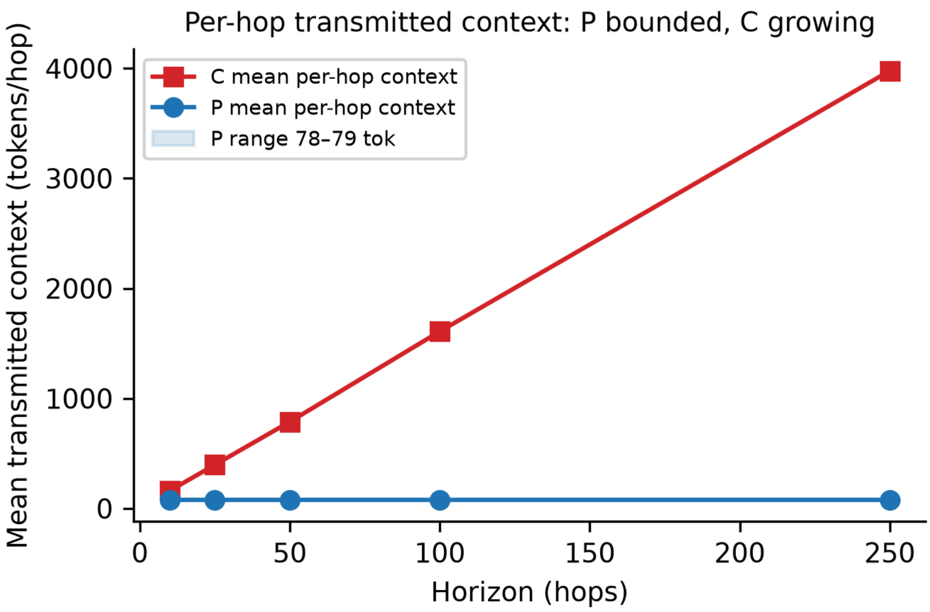

Figure 2: Per-hop transmitted context: P bounded, C growing.

5.5 Scaling behavior

P (packet condition). Approximately linear cumulative growth over the measured horizons:

- Linear fit: `y = 14676.770143x + 4192.197598`, $R^2 = 0.999998$
- Power-law exponent: 0.9946

C (conversational condition). Strongly super-linear cumulative growth:

- Quadratic fit: `y = 568.922736x² + 11268.721360x − 7709.001538`, $R^2 = 0.999999$
- Linear fit R^2: 0.960105 (quadratic dominates)
- Power-law exponent: 1.6905

These are characterizations of measured growth over horizons 10–250. They are not formal proofs of asymptotic $O(N)$ versus $O(N^2)$.

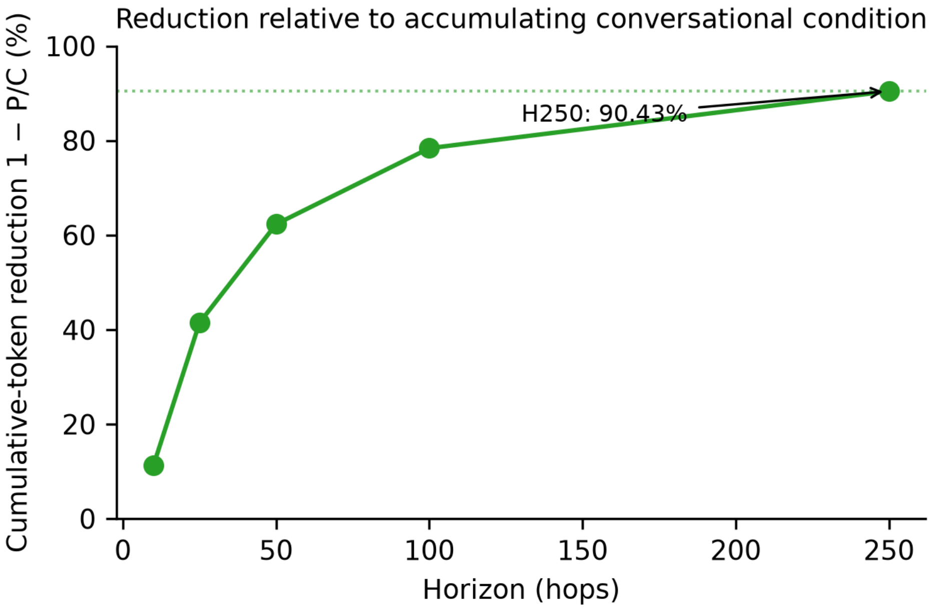

Figure 3: Cumulative-token reduction relative to the accumulating conversational condition by horizon.

5.6 Reliability and state integrity

Two distinct properties are reported separately and must not be conflated:

- **Worker formatting success** — the fraction of worker outputs that parsed into the required structured JSON.
- **State integrity** — the fraction of hops in which the state after the hop equals the state the protocol requires (divergence == 0 and digest match).

A malformed worker response is a worker-output failure; state divergence is an integrity failure. An output can fail to parse while state remains intact (a failed parse is recorded as a non-action with no ledger transition, hence zero divergence), and an output can parse correctly while state diverges (as observed in C at H100).

Aggregate across 26,100 calls:

Cond	Total	PASS	Parse err	Action err	State corr	Divergence rate	Digest continuity
State integrity	P	13,050	99.98%	0.02%	0.00%	0.00%	0.00%
100.00%	100.00%	C	13,050	99.98%	0.01%	0.01%	0.01%
0.02%	99.98%	99.98%					

Table 4: Aggregate reliability across all 26,100 calls.

H250 only:

Cond	Total	PASS	Parse err	Action err	State corr	Divergence rate	Digest continuity
State integrity	P	7,500	99.97%	0.03%	0.00%	0.00%	0.00%
100.00%	100.00%	C	7,500	99.99%	0.01%	0.00%	0.00%
0.00%	100.00%	100.00%					

Table 5: Reliability at H250 only (7,500 P + 7,500 C calls).

Precise statement for the packet condition. PM-1 maintained 100% state integrity and digest continuity across all 13,050 P hops, while 99.98% produced successfully formatted worker responses; the three PARSE_ERROR events produced no state divergence.

5.7 Failure taxonomy

Five non-PASS records across the full 26,100-call dataset:

Horizon	Scenario	Trial	Cond	Hop	Type	Divergence
100	8	2	C	2	ACTION_ERROR (worker action failure)	−1
100	8	2	C	3	STATE_CORRUPTION (persistent state corruption)	−1
250	3	1	C	79	PARSE_ERROR (worker output-format failure)	0
250	3	2	P	102	PARSE_ERROR (worker output-format failure)	0
250	9	1	P	188	PARSE_ERROR (worker output-format failure)	0

Table 6: All five non-PASS records across the 26,100-call dataset.

The ACTION_ERROR and STATE_CORRUPTION occurred under C at H100 (scenario 8, trial 2, hops 2–3) with divergence −1, corresponding to the two expected-vs- actual digest mismatches in the dataset reconciliation. All three H250 parse errors had zero state divergence; none propagated to later hops. The two state-related failures occurred in C; none occurred in P. The distribution is consistent with formatting failures being condition-independent (they occurred in both P and C), while the only state failures occurred in the conversational condition. No causal claim about failure prevention is made; the harness's role is capture and classification.

6. Discussion

The results establish two things. First, PM-1 state survives long sequential handoffs (V0.4a), and the packet condition maintains bounded per-hop context with 100% state integrity and digest continuity (V0.5). Second, the transmission economics of skip-based handoff separate sharply from those of accumulating conversation within this benchmark: approximately linear versus strongly super-linear cumulative growth over the observed horizons, with a 90.43% cumulative-token reduction relative to the accumulating conversational condition at H250.

The result is not simply "JSON is shorter than conversation." The architectural interpretation is that PM-1 changes what information crosses the worker boundary. Token savings and bounded context are consequences of that design, not of a better encoding: the P packet is a compact JSON document, but the boundedness comes from dropping history at the handoff boundary rather than compressing and forwarding it. Compression inherits the input's growth; skip semantics do not.

For long-running agent systems, the implication is that larger context windows provide capacity, while PM-1 reduces unnecessary occupation of that capacity. The benchmark does not measure needle-in-a-haystack accuracy, retrieval quality, or long-context comprehension; those are future experiments. Within this benchmark, the results support the hypothesis that skip-based bounded transmission changes the long-horizon cost structure of handoffs.

The reliability distinction is central. Worker formatting success (99.98% in P) and state integrity (100% in P) are different properties; a formatting failure left state intact, while the only state failures in the dataset occurred in the conversational condition. The results are consistent with protocol integrity being independent of individual worker output quality, but they do not support a claim that PM-1 prevents model failures.

7. Related Work

7.1 Agent memory and persistent state

Memory-augmented LLM agents manage the gap between finite context windows and long-running interaction. MemGPT treats the context window as "main memory" and external storage as "disk," with the model paging data between them via self-directed function calls [Packer et al., 2023]. Generative agents maintain a persistent natural-language memory stream, retrieving relevant memories and synthesizing reflections over long-horizon simulation [Park et al., 2023]. These systems answer a common question: **where should old information live?** PM-1 asks a different question: **what information actually needs to cross the handoff boundary?**

7.2 Retrieval-based memory

Retrieval-augmented generation combines parametric generation with a non-parametric index, retrieving relevant content at generation time [Lewis et al., 2020]. Retrieval injects relevant historical or external content on demand. PM-1 does not retrieve prior conversation; it transmits canonical continuation state. Retrieval-based memory is a potential future-work baseline but was not tested in this benchmark.

7.3 Context compaction and compression

Prompt compression reduces a given prompt to a smaller representation while preserving semantics [Jiang et al., 2023]. Long-context dialogue compression achieves similar ends inside the attention mechanism, caching compacted utterance state rather than full history [Li et al., 2024]. These methods represent the **compression** family that PM-1 explicitly contrasts with **skip**: compression keeps the same accumulated information in a smaller form, which still grows with history; PM-1 does not transmit known or unnecessary history at all. No compaction baseline was tested in this benchmark, and none is claimed.

7.4 Multi-agent handoff and shared state

Multi-agent frameworks coordinate agents through conversational message passing [Wu et al., 2023]. Blackboard architectures provide a shared structured state space through which independent knowledge sources communicate [Hayes-Roth, 1985; Nii, 1986] — the classical precedent for transmitting state rather than interaction history. PM-1 builds on this precedent at the LLM handoff layer: it does not claim that structured state is new; it formalizes a selective transmission boundary for sequential fresh workers and measures the token/context consequences.

7.5 Context-window limitations

Models do not use long contexts robustly: performance is U-shaped with respect to the position of relevant information and degrades as context grows [Liu et al., 2024]. This motivates the architectural alternative of not accumulating long context at all.

[CITATION NEEDED] formal taxonomies of agent handoff and stateful-agent architecture surveys beyond the works above; no verified source was available at the time of writing, and none is fabricated.

8. Limitations

- **Synthetic, deterministic task.** The portfolio task is a vehicle for controlled measurement, not a trading claim.
- **Simple state representation.** Continuation state is a small integer vector; no heterogeneous state was tested.
- **One model, one provider.** deepseek-v4-flash via one endpoint; no cross-model or cross-provider validation.
- **Temperature 0.** Greedy decoding; stochastic-sampling behavior untested.
- **Accumulating conversational baseline.** C is a naive baseline; no compaction, summarization, or retrieval baseline was tested. The measured 90.43% reduction is specifically relative to the accumulating conversational condition.
- **No evolving institutional knowledge.** No decisions, taxonomy, or knowledge in the packet; knowledge transmission and institutional memory are untested.
- **No heterogeneous models or agents.** Cross-model interoperability and heterogeneous-agent swarms are untested.
- **Finite measured horizon.** H10–H250; H500 and H1000 were registered but not executed; no extrapolation.
- **Payload-shaped subset.** Full envelope semantics, field-level delta selection, and intelligent selection were not exercised.
- **Encoding.** Substitution was tested with exactly one alternative (JSON); broader encoding agnosticism is not experimentally demonstrated.
- **Attention behavior.** The benchmark measures transmission/context economics, not model attention or long-context comprehension.

The strongest remaining reviewer objection is: **"C is a strawman; practical systems may use compaction, summarization, retrieval, or external memory."** This is valid and disclosed. The paper demonstrates the advantage of skip-based bounded transmission over the accumulating-transcript condition tested; it does not demonstrate superiority over all practical context-management techniques. A stronger full-venue submission should add those baselines and ideally a second model.

9. Future Work

9.1 The next experiment: evolving institutional truth

The next experimental phase addresses the "integer relay race" criticism directly. Its question:

> Can a bounded handoff packet preserve the CURRENT institutional truth when knowledge changes over time?

The phase must add, relative to V0.5: stable decision IDs; decision supersession; effective ticks; active knowledge selection (the packet carries the currently-active set, not merely the most recent records); taxonomy evolution; detector and validation logic evolution; and fresh workers applying the current institutional truth. Heterogeneous models should be introduced in a later phase, not retroactively claimed. This is a new experiment; it must not be presented as a reinterpretation of V0.5, and the V0.5 paper does not claim to have answered it.

9.2 Research program: the skip taxonomy

A broader research direction organizes skip behavior into categories. These are architectural hypotheses for future work; V0.5 did not validate them.

- **Invariant skip.** Fixed content (e.g., a national anthem) that the receiving worker already possesses does not need retransmission. Known + invariant → skip.
- **Semantic / bounded skip.** Content is variable but the task has an allocated envelope (e.g., a 30-minute keynote); variable content does not imply unlimited budget. This maps naturally to token budgets: instruction budget, evidence budget, reasoning budget, handoff budget, total budget.
- **Process / temporal skip.** A long-running task does not require the agent to carry its entire execution history as active context: work, work, work, meaningful state change, checkpoint, continue. The agent preserves what the work resulted in, not every transient step.

The broader architectural principle under investigation is: **SKIP what is invariant, BOUND what is variable, PERSIST what is durable, RETRIEVE what is needed.** None of this is claimed as a V0.5 result.

9.3 PM-Adapter as future architecture

PM-Adapter is documented separately as a software artifact and technical note. Its potential future architectural role is to adapt heterogeneous domain state into what PM-1 needs: heterogeneous domain state → PM-Adapter (selection / projection / prioritization / budget) → PM-1 → handoff → fresh worker. The corresponding state hierarchy is **S_total** (everything the system knows), **S_active** (currently relevant state), and **S_handoff** (what this particular worker needs), with the hypothesis that **S_handoff** ≪ **S_active** ≪ **S_total** when the domain permits it. This is future architecture, not a V0.5 result. PM-Adapter was not the experimental variable in V0.5 and is not the basis of the V0.5 scientific claim.

10.5 Code and Data Availability

The benchmark implementation, experiment configuration, analysis tooling, and released experimental artifacts are publicly available at https://github.com/rikirinjani/pro-memoria (release v0.5.0). The raw experimental records and archival release are available through Zenodo: https://doi.org/10.5281/zenodo.22005884.

The released artifact contains: the benchmark harness and offline test suite; the V0.5 experiment configuration and status documents (26,100 executed calls across horizons H10, H25, H50, H100, H250; H500 and H1000 registered but NOT executed); raw hop records and checkpoint with a checksum manifest; and the consolidation report. The archive supports offline reproduction and analysis of the reported results. API-backed experimental reproduction requires provider credentials and paid calls; it does not require executing H500 or H1000. The complete 116,100-call registered matrix was not executed.

10. Conclusion

Pro Memoria is a handoff architecture, not a compression codec. Its principle is **skip, don't compress**: known or unnecessary history is not retransmitted, and the receiving worker receives the state required to continue. Across 30,100 completed API-backed handoffs, the architecture maintained 100% state integrity and digest continuity in the packet condition, kept per-hop context bounded at approximately 78–79 tokens, and exhibited approximately linear cumulative token growth while the accumulating-conversation condition grew strongly super-linearly — a 90.43% cumulative-token reduction relative to that condition at H250.

PM-1 demonstrated that repeated fresh-worker handoffs can preserve state integrity while transmitting a bounded continuation packet, and that, in the tested benchmark, this produced increasingly large token savings relative to accumulating conversational history. The architecture is designed to be encoding-agnostic; Morse is the original encoding implementation, and this benchmark operated with a PM-1-shaped JSON packet without changing the architecture or its results.

The results do not show that PM-1 solves agent memory or eliminates context limits; they motivate the next investigation into selective, budgeted, heterogeneous state handoff, and specifically into whether a bounded packet can preserve current institutional truth as knowledge changes over time.

11. References

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[CITATION NEEDED] for formal agent-handoff taxonomies and stateful-agent architecture surveys — not fabricated; to be added from verified sources.